

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-IV (NEW) EXAMINATION – WINTER 2021****Subject Code:3140610****Date:24/12/2021****Subject Name:Complex Variables and Partial Differential Equations****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Prove that $i^i = e^{-(4n+1)\frac{\pi}{2}}$	03
	(b) Show that the function $u=x^2-y^2$ is harmonic and find the corresponding analytic function	04
	(c) State DE MOIVRE'S Theorem and Find the fifth root of unity	07
Q.2	(a) Prove that $\sinh^{-1} x = \log(x + \sqrt{x^2 + 1})$	03
	(b) Show that $f(z) = \frac{\operatorname{Re}(z^2)}{ z }, z \neq 0$ $= 0, z=0$ Is continuous at $z=0$.	04
	(c) State Necessary condition for function to be analytic and Show that neither $f(z)=\bar{z}$ nor $f(z)= z $ is an analytic function.	07
	OR	
	(c) Using Residue Evaluate $\int_0^\infty \frac{x^2 dx}{(x^2+1)(x^2+4)}$.	07
Q.3	(a) Evaluate $\int_c (x^2 - iy^2) dz$ along the parabola $y=2x^2$ from (1,2) to (2,8)	03
	(b) Discuss the convergence of the series. $\sum_{n=0}^\infty \frac{(2n)!(z-3i)^n}{(n!)^2}$	04
	(c) State Cauchy Integral Formula and evaluate $\int_c \frac{z-1}{(z+1)^2(z-2)} dz$, where c is the circle $ z-i =2$	07
	OR	
Q.3	(a) Expand $f(z)=\frac{z-\sin z}{z^2}$ at $z=0$, classify the singular point $z=0$.	03
	(b) Define Mobius Transformation which maps $z=0, i, 1$ into $w=i, -1, \infty$	04
	(c) Determine the poles of the function $f(z)=\frac{z^2}{(z-1)^2(z+2)}$ and residue at each pole. Hence evaluate $\oint_c f(z)$ where c is the circle $ z =3$.	07
Q.4	(a) Solve $z^2(p^2z^2+q^2)=1$	03

- (b) Solve $x(y-z)p+y(z-x)q=z(x-y)$ **04**
 (c) Using method of separation of variable solve $\frac{\partial u}{\partial x}=2\frac{\partial u}{\partial t}+u$, **07**
 where $u(x,0)=6e^{-3x}$

OR

- Q.4** (a) Solve $q=3p^2$ by Charpit method **03**
 (b) Solve $(D^2+10DD'+25D'^2)Z=e^{3x+2y}$ **04**
 (c) A tightly stretched string with fixed end points $x=0$ and $x=l$ is initially in the equilibrium position. If it is set vibrating by giving to each of its points Q , a velocity of $v_0 \sin^3 \frac{\pi x}{l}$. Find the displacement $y(x,t)$. **07**

- Q.5** (a) Form a partial differential equation by eliminating arbitrary function $xyz=\phi(x+y+z)$ **03**
 (b) Classify the following partial differential equation **04**
 1) $2\frac{\partial^2 u}{\partial x^2}+4\frac{\partial^2 u}{\partial x \partial y}+3\frac{\partial^2 u}{\partial y^2}=6$
 2) $\frac{\partial^2 u}{\partial x^2}+4\frac{\partial^2 u}{\partial x \partial y}+4\frac{\partial^2 u}{\partial y^2}=0$

- (c) Find the solution of wave equation $\frac{\partial^2 u}{\partial t^2}=c^2 \frac{\partial^2 u}{\partial x^2}$, **07**
 $0 \leq x \leq L$ satisfying the condition $u(0,t)=u(L,t)=0$
 $u_t(x,0)=0, u(x,0)=\frac{\pi x}{L}, 0 \leq x \leq L$

OR

- Q.5** (a) Solve $\frac{\partial^2 z}{\partial x^2}=z$ **03**
 (b) Find the solution of $u_t=c^2 u_{xx}$ together with the initial and boundary conditions $u(0,t)=u(l,t)=0$ for all $t \geq 0$ and $u(x,0)=\sin \frac{\pi x}{l}, 0 \leq x \leq l$ **04**
 (c) A rod 30 cm long has its end A and B Kept at 20° and 80° **07**
 c respectively until steady state condition prevail. The temperature at each end is then suddenly reduced to 0° kept so. Find the resulting temperature function $u(x,t)$ from the end A.
